



KULDEEP VARMA

Master of Medical Science and Technology(M.M.S.T.)

Medical doctor and AI researcher

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EDUCATION

Degree	Institute/Board	Year
M.M.S.T.	Indian Institute of Technology, Kharagpur	2023-2026
M.B.B.S.	Govt. Medical College, Nagpur	2011-2017

SUMMARY

Medical doctor with 5 years of clinical experience and an AI researcher passionate about combining medicine, computer vision, and machine learning to solve real-world healthcare problems. Proficient in medical imaging, bioinformatics, and data-efficient AI methods, with a strong interest in clinically impactful and interdisciplinary research.

EXPERIENCE

- Kharagpur Learning, Imaging and Visualization (KLIV) research group**
Research Intern May 2025 – May 2026
IIT Kharagpur, India
 - Developed a novel cold-start graph-based transductive label propagation framework for surgical videos using frozen self-supervised (SSL) representations, enabling training-free annotation propagation from minimal labels without requiring domain-specific training.
 - Benchmarked 17 SSL encoders across ResNet and ViT architectures using two label propagation methods kNN and LLGC. LLGC achieves mAP scores of 0.51 to 0.91 under 1% to 20% labeled supervision, consistently outperforming kNN with a maximum relative gain of 7.7% at 1% supervision, highlighting the effectiveness of graph-based propagation.
 - Demonstrated that SSL propagation performance strongly correlates with Krippendorff's α agreement ($r = 0.98$) and embedding geometry separability ($r = 0.62$), suggesting that semantic consistency and well-structured embeddings improve graph-based label propagation.
 - Authoring a research manuscript for submission to a Tier-A computer vision conference.
- Primary Health Centers**
Medical Officer Feb 2018 – Jul 2023
Maharashtra & Madhya Pradesh
 - Led healthcare teams delivering primary and emergency care while overseeing implementation of government health programs. Managed clinical operations focusing on NCDs, maternal and child health, and mental health services.
- B.C. Roy Hospital**
Medical Officer Jul 2023 – jun 2026
IIT Kharagpur, West Bengal
 - Balanced hospital duties with M.M.S.T. coursework while helping implement student healthcare programs. Diagnosed student health conditions and delivered clinical care within the IIT Kharagpur campus health system.

PROJECTS

- In-domain SimCLR for Surgical Video Annotation**
Under Prof. Debdoot Sheet Jan 2025 – Apr 2025
 - Trained an in-domain SimCLR self-supervised model on the Cholec80 surgical video dataset. Performed surgical instrument annotation using cosine similarity-based label propagation with hard thresholding.
- Bioinformatics Analysis of Interstitial Lung Disease**
Under Prof. Koel Chaudhury Sept 2024 – Dec 2024
 - Conducted network-based bioinformatics analysis to identify key genes associated with Interstitial Lung Disease (ILD) and chronic hypersensitivity pneumonitis. Identified 12 potential target genes using CytoHubba (MCC and Bottleneck metrics) and enrichment analysis, highlighting their role in lung fibrosis pathways.

TECHNICAL SKILLS

- Programming Languages:** Python, MATLAB, SPSS
- Tools & Libraries:** PyTorch, Torchvision, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib, Scikit-learn, MediaPipe, Hugging Face Transformers, MONAI, SimpleITK

RELEVANT COURSEWORK

- Imaging:** Digital Image Processing and Applications, Medical Imaging, Biomedical Imaging Informatics, Molecular Imaging
- Engineering & Quantitative:** Robotics, Bio-statistics and Quantitative Techniques in Medicine and Biology, CAD (ABAQUS), Fundamentals of Medical Instrumentation, Evidence-Based Medicine, Intro. to Intellectual Property Rights

ACHIEVEMENTS

- GATE:** All India Rank 82 (Biomedical Engineering)
- Sports:** University Chess Championship – Runner-up

CERTIFICATIONS

- Computer Vision:** OpenCV Bootcamp, PyTorch Bootcamp, Vision-Language Model Bootcamp
- Finance:** Summer of Quant (Quantitative Finance)